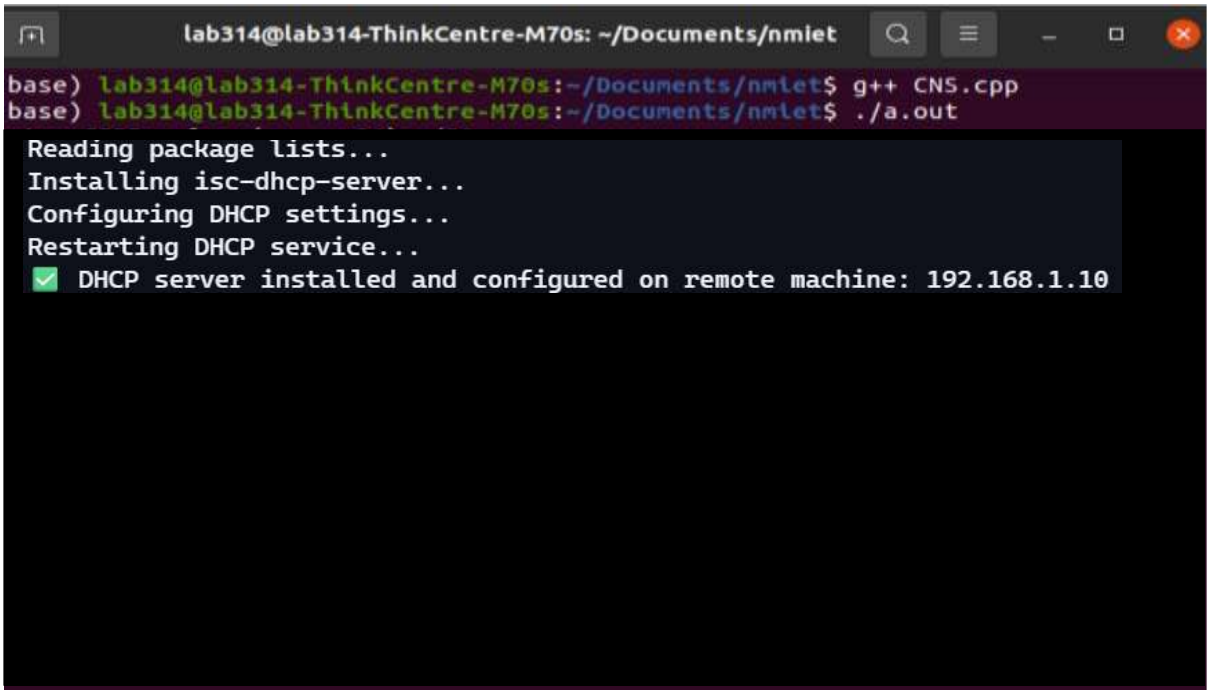
Practical - 13

Installing and configure DHCP server and write a program to install the software on remote machine.

```
#include <iostream>

#include <cstdlib>

#include <string>


using namespace std;


int main() {

   string remoteIP = "192.168.1.10";     // Remote machine IP

   string username = "user";           // Remote username

   string sshPrefix = "ssh " + username + "@" + remoteIP + " ";


   // Step 1: Update and install DHCP server

   string installCmd = sshPrefix + "\"sudo apt update && sudo apt install -y isc-dhcp-server\"";

   system(installCmd.c_str());


   // Step 2: Configure DHCP interface

   string ifaceCmd = sshPrefix + "\"echo 'INTERFACESv4=\\\"eth0\\\"' | sudo tee /etc/default/isc-
dhcp-server\"";

   system(ifaceCmd.c_str());


   // Step 3: Configure DHCP settings

   string dhcpConfig =

      "echo 'default-lease-time 600;\\n"

      "max-lease-time 7200;\\n"

      "subnet 192.168.1.0 netmask 255.255.255.0 {\\n"

      "  range 192.168.1.100 192.168.1.200;\\n"

      "  option routers 192.168.1.1;\\n"

      "  option domain-name-servers 8.8.8.8, 8.8.4.4;\\n"

      "}' | sudo tee /etc/dhcp/dhcpd.conf";
```

```
    string configCmd = sshPrefix + "\"" + dhcpConfig + "\"";

    system(configCmd.c_str());


    // Step 4: Restart and enable DHCP service

    string restartCmd = sshPrefix + "\"sudo systemctl restart isc-dhcp-server && sudo systemctl enable
isc-dhcp-server\"";

    system(restartCmd.c_str());


    cout << " ✅ DHCP server installed and configured on remote machine: " << remoteIP << endl;

    return 0;

}
```

// OUTPUT